



INFRASTRUCTURE PLANNING REPORT

Vida Hosting & Scaling Cost Analysis

Render, MongoDB Atlas, Cloudflare R2, Android distribution and domain costs

\$16-\$18/month beta baseline → **\$710-\$902/month** at 100,000 MAU

Prepared for Vida

20 July 2026 | Singapore

Executive summary

PLANNING RECOMMENDATION Budget about \$20/month for a controlled beta, \$85/month for a dependable small production launch, \$110/month near 10,000 MAU, and approximately \$800/month near 100,000 MAU.

The number of registered accounts alone does not drive hosting cost. This report uses monthly active users (MAU), because requests, bandwidth, stored content and peak concurrency are what create spend.

Cost outlook at a glance

Monthly active users	Estimated monthly	Estimated annual	Practical interpretation
100	\$16-\$18	\$192-\$216	Lean beta. Shared database and one small API instance.
1,000	\$17-\$45	\$204-\$540	Controlled beta or early launch. Upgrade if uptime is business-critical.
10,000	\$103-\$114	\$1,236-\$1,368	Standard API plus dedicated Atlas M10.
100,000	\$710-\$902	\$8,520-\$10,824	Pro workspace, one or two Pro API instances and Atlas M30.

Table 1. Core infrastructure estimate; USD before tax. Optional maps and third-party messaging are excluded.

What matters most

- **Media architecture is cost-efficient.** Images upload and download through Cloudflare R2 rather than being proxied through Render.
- **Database reliability creates the first meaningful fixed-cost step.** Atlas Flex is inexpensive, but MongoDB positions it for development and testing; an M10 is the more credible production floor.
- **Bandwidth becomes material at scale.** Under Render's new Hobby allowance, 100 GB/month costs about \$14 in overage and 1 TB/month costs roughly \$146 on Pro.
- **Map services are the main unpriced dependency.** The current public OpenStreetMap tile and Photon endpoints provide no commercial SLA and can throttle heavy use.

Detailed monthly cost breakdown

The figures below represent a reasonable capacity plan for the current architecture. They are budget estimates rather than vendor quotes; actual spend varies with region, traffic shape, query efficiency and retention.

Cost component	100 MAU	1,000 MAU	10,000 MAU	100,000 MAU
Render API compute	\$7	\$7	\$25	\$85-\$170
Render workspace plan	\$0	\$0	\$0	\$25
Render static frontends	\$0	\$0	\$0	\$0
Render bandwidth	\$0	~\$0.75	~\$14	~\$146
MongoDB Atlas	\$8	\$8-\$30	~\$57	~\$394
Atlas backup / transfer reserve	\$0	\$0-\$5	\$5-\$15	\$50-\$150
Cloudflare R2 media	\$0	\$0	~\$1	\$9-\$15
Domain registration	\$1-\$2	\$1-\$2	\$1-\$2	\$1-\$2
Estimated total	\$16-\$18	\$17-\$45	\$103-\$114	\$710-\$902

Table 2. Estimated recurring monthly spend. The Google Play developer registration fee is a separate one-time cost.

PRODUCTION FLOOR If uptime matters from day one, use Render Standard plus Atlas M10 even at 100-1,000 MAU. That produces a safer baseline of roughly \$83-\$86 per month.

Assumptions and additional costs

Estimate basis

- Users means monthly active users (MAU), not total registered accounts.
- The model assumes about 300 API calls and 10 MB of Render-delivered data per active user each month.
- Images are delivered directly through Cloudflare R2, keeping media away from Render bandwidth.
- Prices are in USD, before tax, and were checked on 20 July 2026.

Other costs to plan for

- Google Play developer registration: \$25 one time.
- Domain registration: approximately \$10-\$20 per year; Render-managed HTTPS is free.
- Paid maps/geocoding: not included. The current public OpenStreetMap and Photon endpoints may not suit large-scale production use.
- Email, SMS, commercial monitoring, paid analytics and staffing are not included.

SIMPLE PLANNING BUDGET Use \$20/month for beta, \$85/month for a dependable small launch, \$110/month near 10,000 MAU, and about \$800/month near 100,000 MAU.

Official pricing sources

1. [Render pricing](#)
2. [Render workspace plan changes](#)
3. [MongoDB Atlas pricing](#)
4. [Cloudflare R2 pricing](#)
5. [Google Play registration](#)
6. [OpenStreetMap tile policy](#)